

Indirect Robot Manipulation using Eye Gazing and Head Movement For Future of Work in Mixed Reality

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ABSTRACT

Most robot manipulations are guided by the worker's both hands using a teach-pendant or other gadgets. However, manipulating the robot is more difficult when both hands are occupied, which requires a new manipulation method. This study proposes a novel indirect robot interaction and manipulation method using eye gaze and head movement in mixed reality. The proposed approach has been implemented to provide hands-free robot interaction for pick-and-place tasks. The results show a promising direction for the future of work in human-robot collaboration.

Keywords: Eye gazing, head movement, indirect robot manipulation, mixed reality.

Index Terms: H.5.2 [Information Interfaces and Presentation]: User Interfaces—Eye Gazing and Head Movement-based Hands-free Interaction; I.2.9 [Artificial Intelligence]: Robotics—Indirect Robot Manipulation; I.3.8 [Computer Graphics]: Applications—Mixed Reality

1 INTRODUCTION

A smart factory is considered one of the most important future directions in the manufacturing industry with the help of information and communication technology, such as artificial intelligence, virtual and augmented reality, and the Internet of Things, which can improve productivity and product quality. Although human-robot collaboration (HRC) can effectively assist complex manufacturing tasks, new interfaces and interactions are required for more safe and effective robot manipulation to improve safety of HRC.

Augmented and mixed reality (AR/MR) can assist the worker in effectively conducting manufacturing tasks [1]. AR/MR can visualize various 3D information for interacting with robots intuitively and naturally. In particular, MR can provide a hybrid of physical and virtual worlds by providing spatial awareness using real-time scanning of the physical environment around the user. MR has also been applied to HRC [2]. Most robot manipulations

were conducted by interacting with the robot using interfaces that utilize worker's both hands, such as teach-pendant and mouse and keyboard. However, to hold tools and devices while performing manufacturing tasks, many workers cannot use both hands. Thus, there is a safety issue since the worker should directly and closely interact with robots and conduct their tasks.

This study proposes a new indirect and hands-free interaction for robot manipulation in MR when workers cannot use both hands. The proposed approach uses eye gazing and head movement-based interaction through smart MR glasses with multiple built-in sensors, such as gyroscopic, accelerometer, and eye tracker. Indirect robot manipulation is conducted by matching the virtual object onto the physical object. When both objects are matched using eye gazing and head movement, the configuration and pose of the virtual objects are used to manipulate the real robot. The proposed approach provide more safe and effective human-robot interaction without hand manipulation. Thus, it offers another perspective of user-centric task assistance in MR.

2 EYE GAZING AND HEAD GESTURE-BASED INDIRECT ROBOT MANIPULATION

For hands-free interaction in MR, eye gazing is used for coarse interaction and head movement is used for fine interaction. Eye gazing is less accurate but faster than head movement, while head movement is more stable and accurate. We use eye gazing for searching objects in the user interface (UI) and head movement for selecting UIs and manipulating 3D virtual objects using pitch, yaw, and roll-based head gestures [3]. Interaction with virtual objects and robots in MR using eye gazing and head movement is shown in Figure 1.

For indirect robot manipulation in MR, virtual-physical object matching is conducted through eye gazing and head movement using smart MR glasses. We assume that it is possible to perform faster and more safe indirect robot manipulation in a hands-free situation by matching the pose of the virtual object to its corresponding physical object rather than directly manipulating the robot end-effector. To minimize eye gazing and head movement for virtual-physical matching, deep learning-based object detection, RetinaNet [4], is applied to find the initial pose of physical objects in MR.

Firstly, images are captured in smart MR glasses and they are used for detecting 2D bounding boxes using RetinaNet. Then, ray-casting is used to map the detected bounding boxes on the reconstructed 3D MR environment [5]. Through initial pose detection and estimation, it is possible to perform 3D manipulation more efficiently for fine-tuning and removing tedious initial matching between virtual and physical objects.

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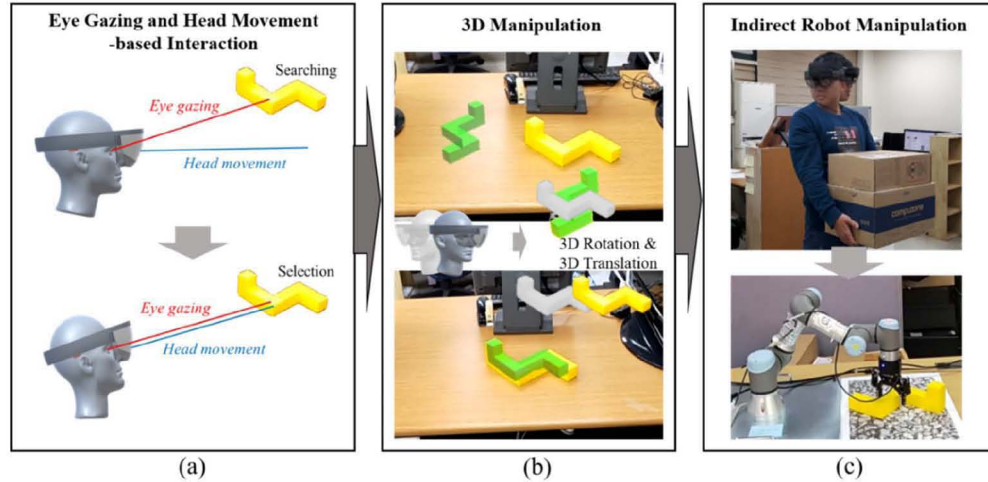


Figure 1: Overview of the proposed eye gazing and head movement-based indirect robot manipulation: (a) searching for an object using eye gazing (red line) and selecting the object using head movement (blue line), (b) manipulating the 3D virtual object (green) to match onto a physical object (yellow) using head movement-based 3D rotation and translation, (c) interacting with a collaborative robot using the proposed indirect robot manipulation when the user cannot use their hands.

The implementation results of the proposed approach are shown in Figure 2. After initial 3D pose estimation of a physical object by applying deep learning-based object detection, the user wearing smart MR glasses manipulates the virtual object to match it exactly onto the physical object through eye gazing and head movement-based interaction. When virtual-physical matching is finished, the inverse kinematics of the virtual robot is calculated based on the pose of the virtual object. Finally, the inverse kinematic data are sent to the real robot which performs pick-place tasks with the physical object. Furthermore, the synchronized virtual robot can help the user perform the robot manipulation more intuitively because the user can preview the final pose and movement before executing the real robot.

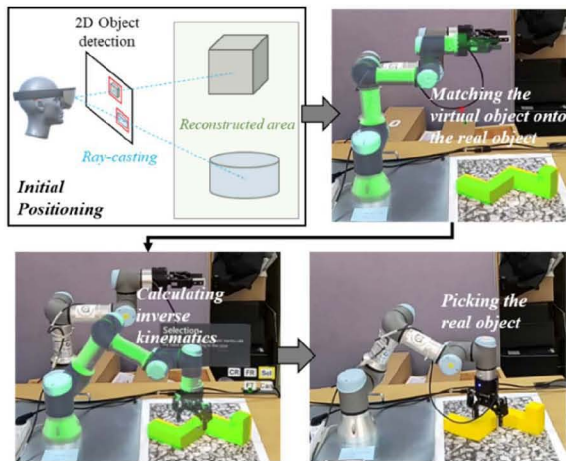


Figure 2: Robot manipulation using hands-free interaction in MR.

3 CONCLUSION

This study proposed a new indirect robot manipulation using eye gazing and head movement in MR environments when the worker cannot use both hands. Eye gazing is used for coarse interaction, such as searching objects and UIs, and head movement for fine interaction, such as selecting UIs and manipulating 3D objects.

The proposed approach can assist the worker in establishing a safe interaction with the robot when a safety accident may occur during HRC. To minimize tedious and error-prone interactions with the robot, deep learning-based object detection is applied to find the initial pose of the virtual object in MR. Virtual-physical matching can provide a safe indirect robot manipulation compared to conventional direct robot manipulation. We will extend the proposed approach to various interactions and collaboration tasks among multiple workers and robots.

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